

Practical 1: WAP to implement connection between client and server using socket programming

EchoClient.java

```
import java.io.*;
import java.net.*;

public class EchoClient {

    public static void main(String[] args) {
        String host = "localhost";
        int port = 3333;

        try {
            Socket socket = new Socket(host,
                                      port);

            BufferedReader userIn = new
                BufferedReader(new InputStreamReader(
                    socket.getInputStream()));

            PrintWriter out = new PrintWriter(
                socket.getOutputStream(), true);
        } {
            System.out.println("Connected to Server");
            Type message, 'stop' to end.");
            String userLine;
```



```
while (CuserLine = userIn.readLine()  
      != null) {  
    out.println(CuserLine);  
    if ("stop".equalsIgnoreCase(  
        CuserLine.trim())) break;  
    String reply = in.readLine();  
    System.out.println("Server replied:  
        + reply);  
}  
catch (IOException e) {  
    e.printStackTrace();  
    System.out.println("Client terminated");  
}
```


EchoServer.java

```
import java.io.*;
import java.net.*;

public class EchoServer {

    public static void main(String[] args)
    {
        int port = 3333;

        System.out.println("Server starting
        on port " + port);

        try {
            ServerSocket serverSocket
            = new ServerSocket(port);

            Socket clientSocket = serverSocket.accept();

            System.out.println("Client connected
            from " + clientSocket.getRemote
            Socket Address());

            try {
                BufferedReader in = new
                BufferedReader(new
                InputStreamReader(clientSocket.
                getInputStream()));

                PrintWriter out = new PrintWriter(
                clientSocket.getOutputStream(),
                true);
            }
        }
    }
}
```



```
{  
    String line;  
    while((line = in.readLine()) != null){  
        if ("stop".equalsIgnoreCase(line.trim()))  
            break;  
        System.out.println("Received from  
client : " + line);  
        out.println("Echo: " + line);  
    }  
    System.out.println("Closing  
connection with client.");  
} catch (IOException e) {  
    e.printStackTrace();  
    System.out.println("Server shutting  
down!");  
}
```